



OSCILLATIONS, WAVES AND OPTICS

PHYSICS FOR CHEMISTRY STUDENTS

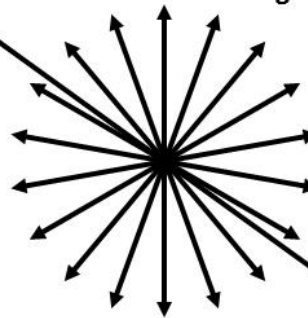
Ye Yang
Xiamen University

4.7 POLARIZATION

Non-polarized



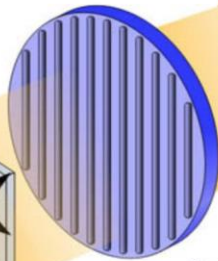
Light waves Vibrate in any direction



Beam of light Propagation

Light waves vibrate
in any direction

Polarized light

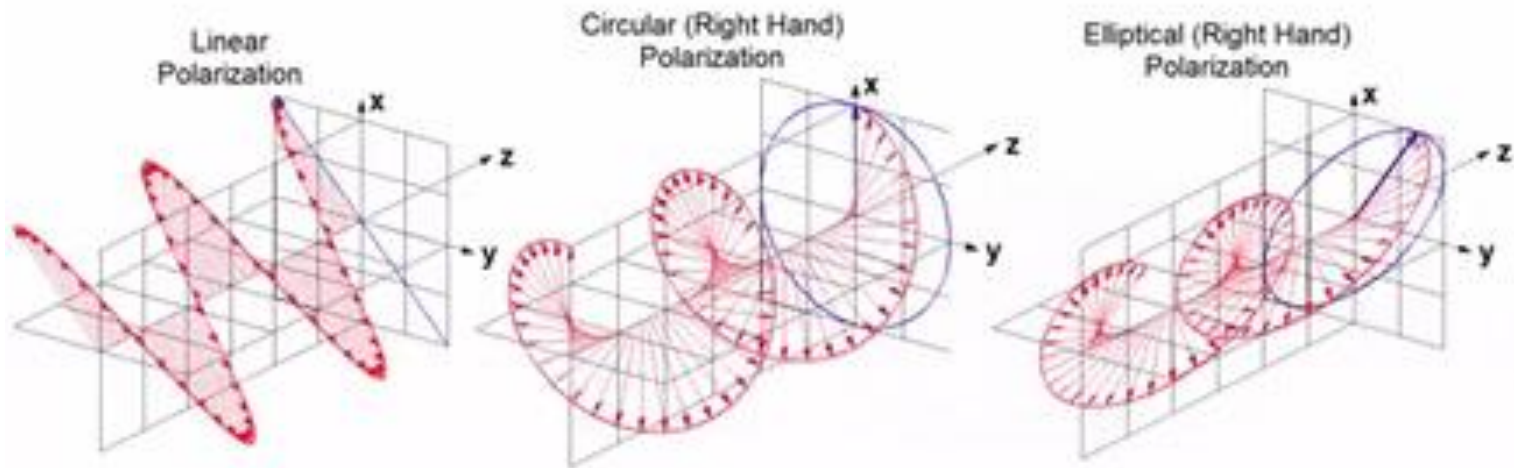


Polarizing filter

Light source



4.7 POLARIZATION



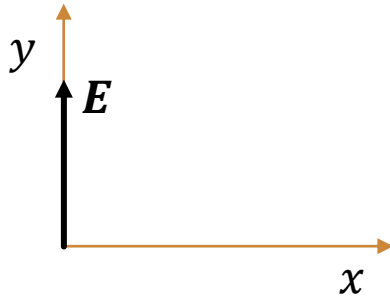
4.7 POLARIZATION

$$\varphi_x - \varphi_y = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \dots$$

$$E_x = E_{0x} \cos(kz - \omega t + \varphi_x)$$

$$E_y = E_{0y} \cos(kz - \omega t + \varphi_y)$$

$$t = t_0$$



$$x: \quad kz_0 - \omega t + \varphi_x = \frac{\pi}{2} \quad E_x = 0$$

$$y: \quad kz_0 - \omega t + \varphi_y = 0 \quad E_y = 1$$

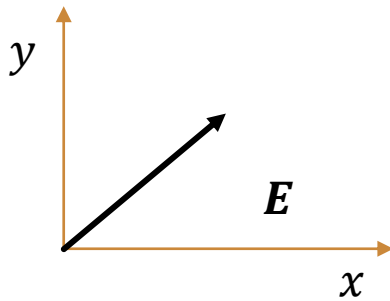
4.7 POLARIZATION

$$\varphi_x - \varphi_y = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \dots$$

$$E_x = E_{0x} \cos(kz - \omega t + \varphi_x)$$

$$E_y = E_{0y} \cos(kz - \omega t + \varphi_y)$$

$$t = t_0 + \frac{T}{8}$$



$$x: \quad kz_0 - \omega t + \varphi_x = \frac{\pi}{2} - \omega \frac{T}{8} = \frac{\pi}{4} \quad E_x = \sqrt{2}/2$$

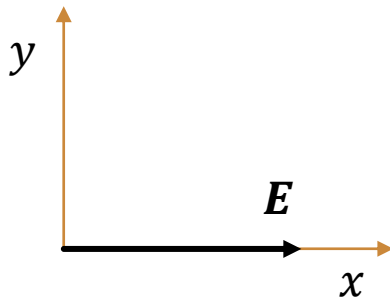
$$y: \quad kz_0 - \omega t + \varphi_y = -\omega \frac{T}{8} = -\frac{\pi}{4} \quad E_y = \sqrt{2}/2$$

4.7 POLARIZATION

$$\varphi_x - \varphi_y = \frac{\pi}{2}, \frac{3\pi}{2}, \frac{5\pi}{2} \dots$$

$$E_x = E_{0x} \cos(kz - \omega t + \varphi_x)$$

$$E_y = E_{0y} \cos(kz - \omega t + \varphi_y)$$



$$t = t_0 + \frac{T}{4}$$

$$x: \quad kz_0 - \omega t + \varphi_x = \frac{\pi}{2} - \omega \frac{T}{4} = 0 \quad E_x = 1$$

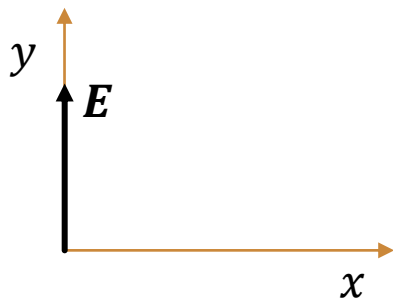
$$y: \quad kz_0 - \omega t + \varphi_y = -\omega \frac{T}{4} = -\frac{\pi}{2} \quad E_y = 0$$

4.7 POLARIZATION

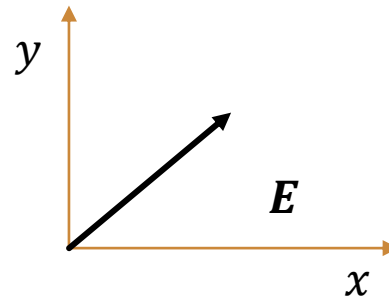
$$\varphi_x - \varphi_y = \frac{\pi}{2}$$

$$E_x = E_{0x}e^{ikz-i\omega t+i\varphi_x}$$

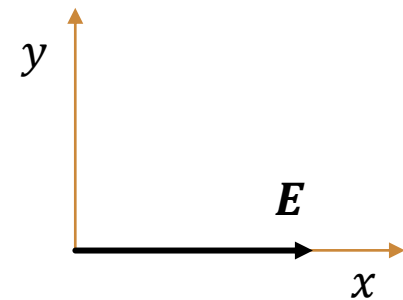
$$E_y = E_{0y}e^{ikz-i\omega t+i\varphi_y}$$



$$t = t_0$$

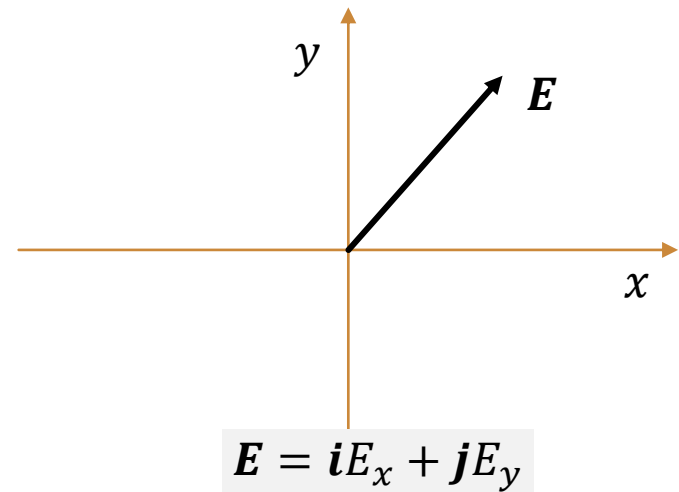
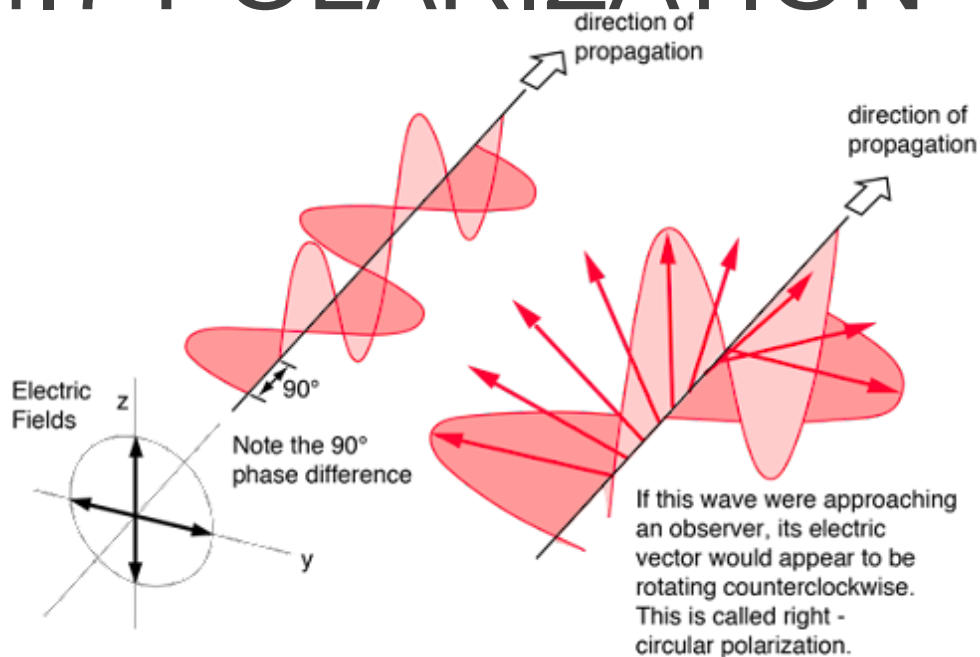


$$t = t_0 + \frac{T}{8}$$



$$t = t_0 + \frac{T}{4}$$

4.7 POLARIZATION



$$E_x = E_{0x}e^{ikz-i\omega t+i\varphi_x} \quad E_y = E_{0y}e^{ikz-i\omega t+i\varphi_y}$$

$$\tilde{E}_0 = iE_{0x}e^{i\varphi_x} + jE_{0y}e^{i\varphi_y}$$

$$E = \tilde{E}_0 e^{ikz-i\omega t}$$

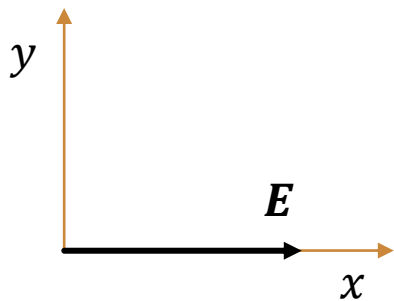
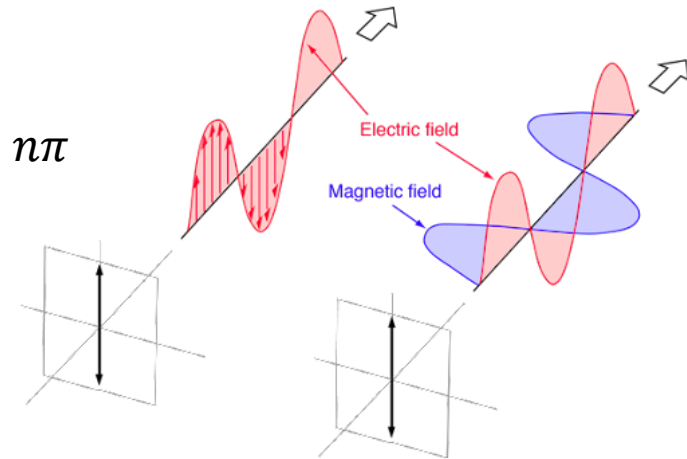
Jones vector:

$$\tilde{E}_0 = \begin{pmatrix} E_{0x}e^{i\varphi_x} \\ E_{0y}e^{i\varphi_y} \end{pmatrix}$$

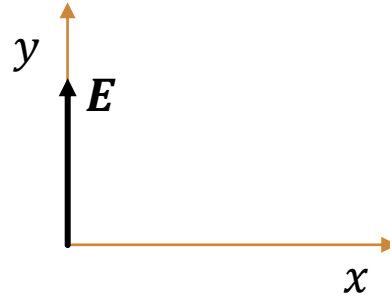
4.7 POLARIZATION

Linear polarization: $\tilde{\mathbf{E}}_0 = \begin{pmatrix} E_{0x}e^{i\varphi_x} \\ E_{0y}e^{i\varphi_y} \end{pmatrix} \quad \varphi_x - \varphi_y = n\pi$

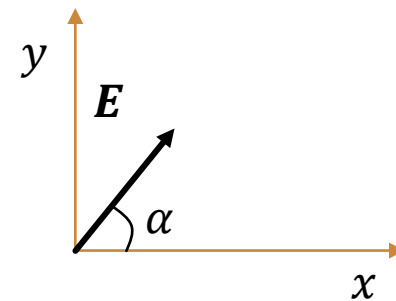
$$\tilde{\mathbf{E}}_0 = e^{i\varphi_y} \begin{pmatrix} E_{0x}e^{i(\varphi_x - \varphi_y)} \\ E_{0y} \end{pmatrix} = e^{i\varphi_y} \begin{pmatrix} \pm E_{0x} \\ E_{0y} \end{pmatrix}$$



$$\tilde{\mathbf{E}}_0 = \begin{pmatrix} E_{0x} \\ 0 \end{pmatrix} = A \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$



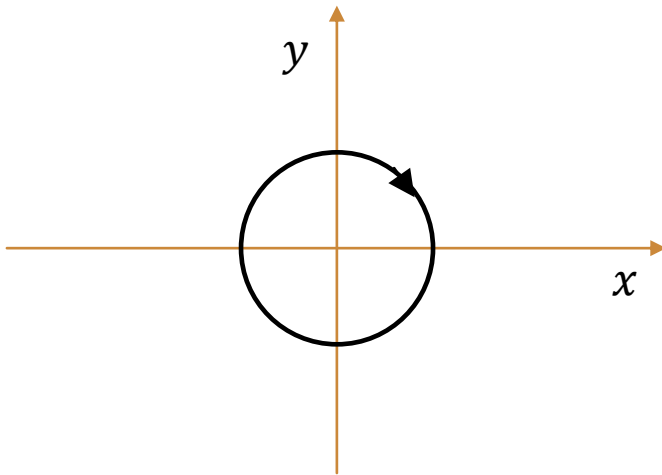
$$\tilde{\mathbf{E}}_0 = A \begin{pmatrix} 0 \\ 1 \end{pmatrix}$$



$$\tilde{\mathbf{E}}_0 = A \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$$

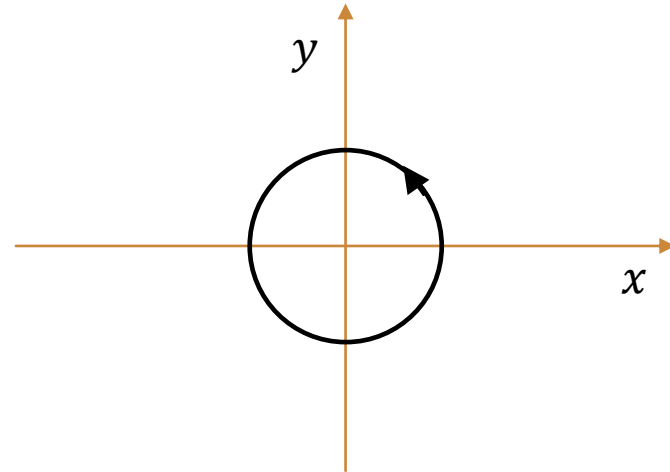
4.7 POLARIZATION

$$\varphi_x - \varphi_y = \frac{\pi}{2}$$



$$E_{0x} = E_{0y}$$

$$\varphi_y - \varphi_x = \frac{\pi}{2}$$



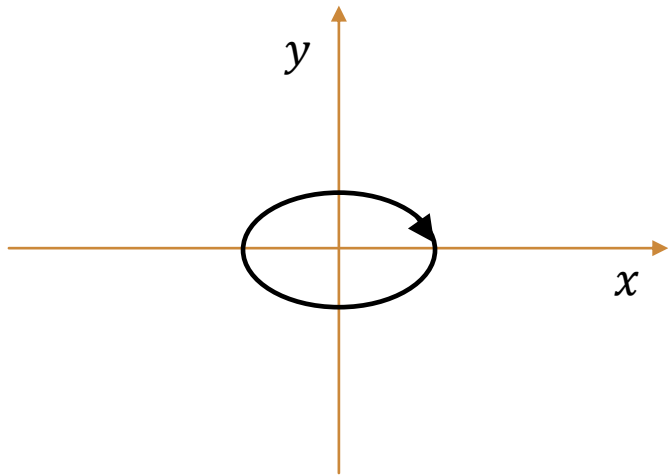
Jones vector:

$$\tilde{\mathbf{E}}_0 = \begin{pmatrix} E_{0x}e^{i\varphi_x} \\ E_{0y}e^{i\varphi_y} \end{pmatrix} = A \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

$$\tilde{\mathbf{E}}_0 = \begin{pmatrix} E_{0x}e^{i\varphi_x} \\ E_{0y}e^{i\varphi_y} \end{pmatrix} = A \begin{pmatrix} 1 \\ i \end{pmatrix}$$

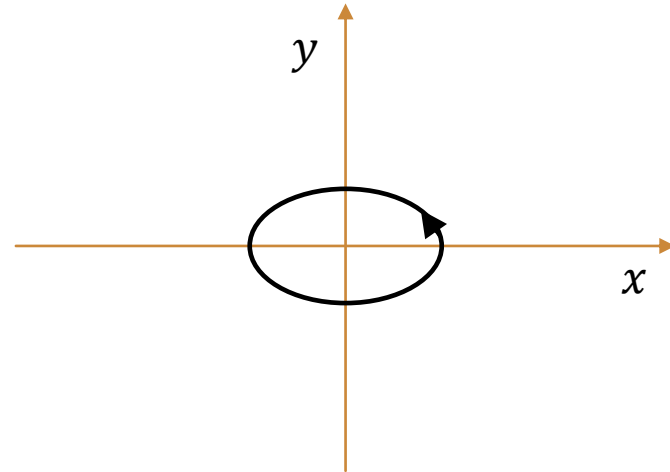
4.7 POLARIZATION

$$\varphi_x - \varphi_y = \frac{\pi}{2}$$



$$E_{0x} > E_{0y}$$

$$\varphi_y - \varphi_x = \frac{\pi}{2}$$



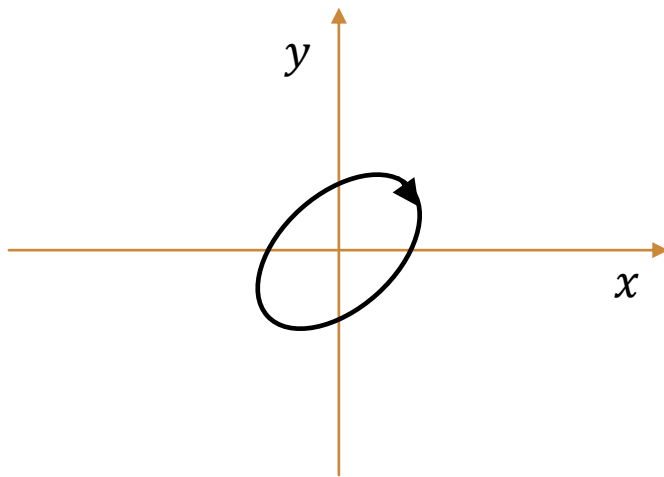
Jones vector:

$$\tilde{\mathbf{E}}_0 = \begin{pmatrix} E_{0x}e^{i\varphi_x} \\ E_{0y}e^{i\varphi_y} \end{pmatrix} = \begin{pmatrix} A \\ -iB \end{pmatrix}$$

$$\tilde{\mathbf{E}}_0 = \begin{pmatrix} E_{0x}e^{i\varphi_x} \\ E_{0y}e^{i\varphi_y} \end{pmatrix} = \begin{pmatrix} A \\ iB \end{pmatrix}$$

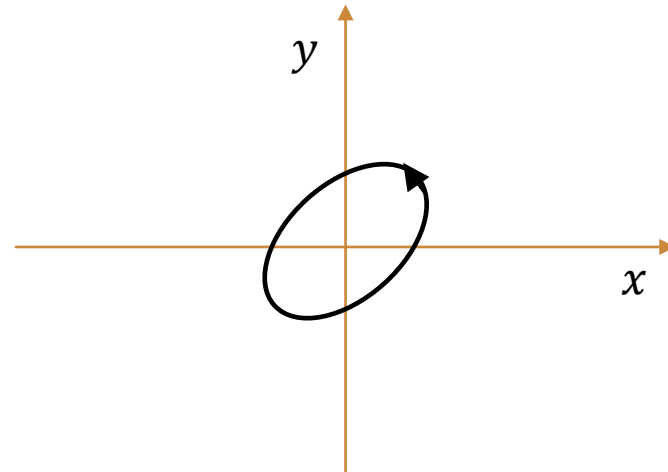
4.7 POLARIZATION

$$\varphi_x - \varphi_y = \theta$$



$$E_{0x} > E_{0y}$$

$$\varphi_y - \varphi_x = \theta$$



Jones vector:

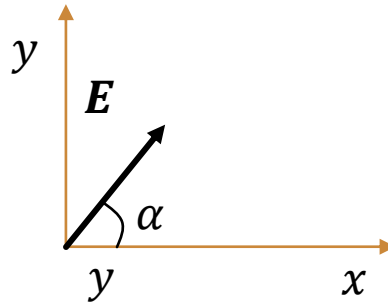
$$\tilde{\mathbf{E}}_0 = \begin{pmatrix} E_{0x}e^{i\varphi_x} \\ E_{0y}e^{i\varphi_y} \end{pmatrix} = \begin{pmatrix} A \\ B - iC \end{pmatrix}$$

$$\tilde{\mathbf{E}}_0 = \begin{pmatrix} E_{0x}e^{i\varphi_x} \\ E_{0y}e^{i\varphi_y} \end{pmatrix} = \begin{pmatrix} A \\ B + iC \end{pmatrix}$$

4.7 POLARIZATION

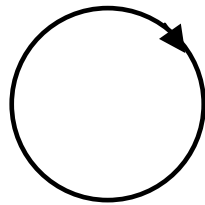
Summary:

linear

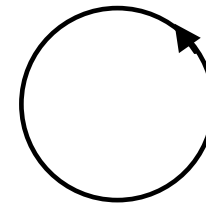


$$\tilde{E}_0 = \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix}$$

circular

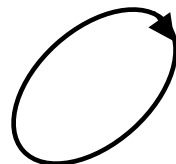


$$\tilde{E}_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$



$$\tilde{E}_0 = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

Elliptical

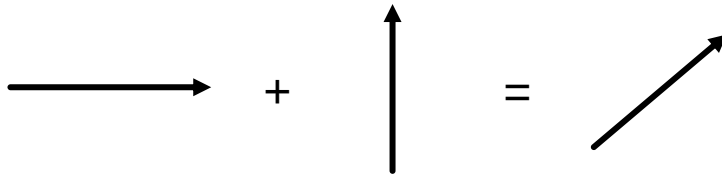


$$\tilde{E}_0 = \frac{1}{\sqrt{A^2 + B^2 + C^2}} \begin{pmatrix} A \\ B - iC \end{pmatrix}$$

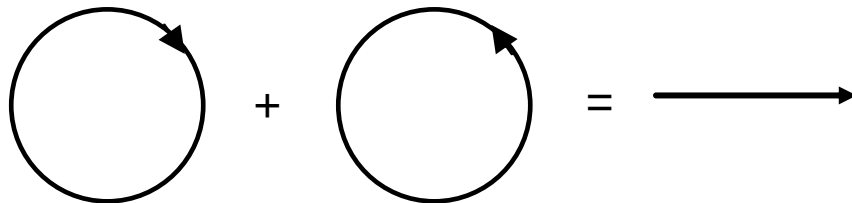
4.7 POLARIZATION

Superposition of polarized modes

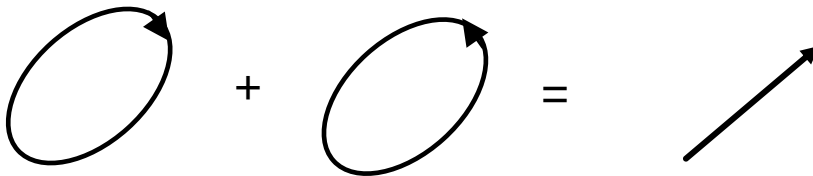
Intensity and polarization mode



$$\begin{pmatrix} 1 \\ 0 \end{pmatrix} + \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 1 \\ 1 \end{pmatrix}$$



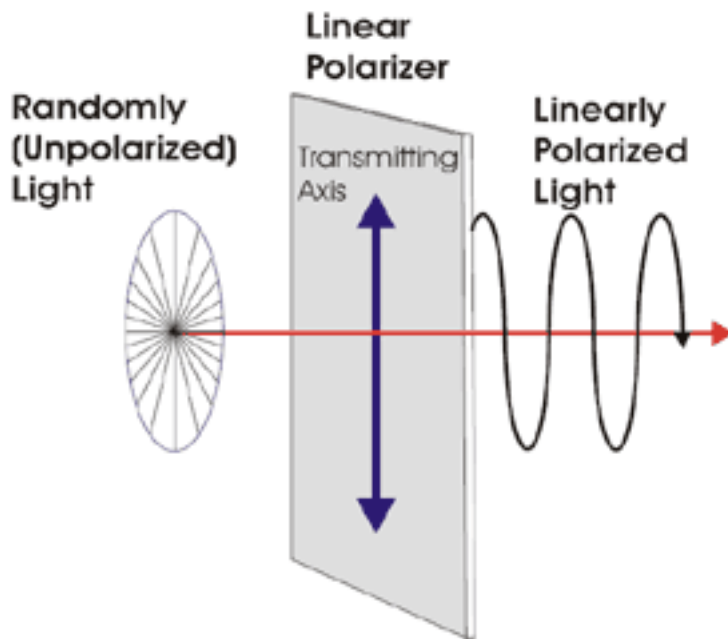
$$\frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix} + \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 2 \\ 0 \end{pmatrix}$$



$$\frac{1}{\sqrt{A^2 + B^2 + C^2}} \left[\begin{pmatrix} A \\ B - iC \end{pmatrix} + \begin{pmatrix} A \\ B + iC \end{pmatrix} \right] = \frac{1}{\sqrt{A^2 + B^2 + C^2}} \begin{pmatrix} 2A \\ 2B \end{pmatrix}$$

4.7 POLARIZATION

Matrix representation of polarizer operation



Vertical linear polarizer

$$M = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}$$

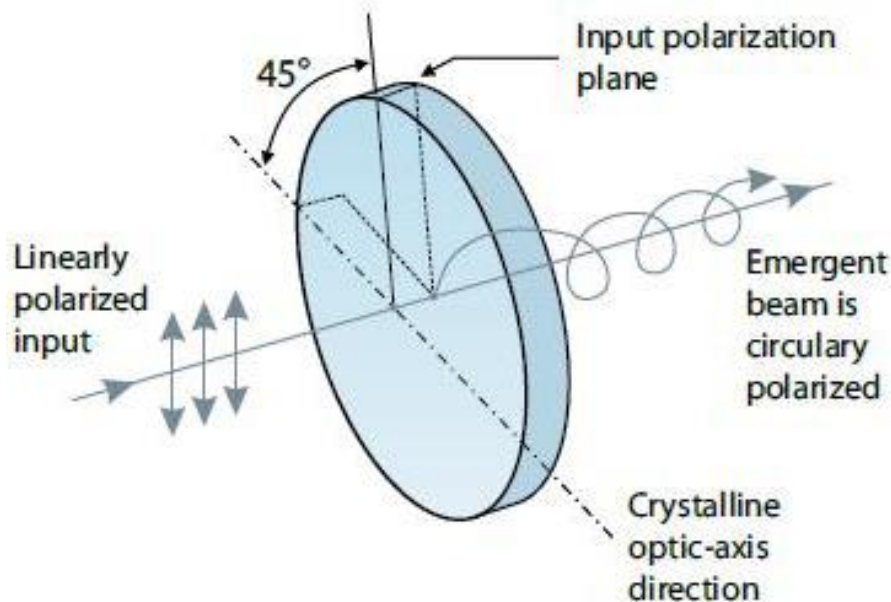
Horizontal polarized light

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 0 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix} \begin{pmatrix} \cos \alpha \\ \sin \alpha \end{pmatrix} = \begin{pmatrix} 0 \\ \sin \alpha \end{pmatrix}$$

4.7 POLARIZATION

Matrix representation of polarizer operation



Performance of Quarter ($\lambda/4$) Waveplate

Quarter waveplate

$$M = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix}$$

Retard vertical axis

$$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} \begin{pmatrix} 1 \\ 0 \end{pmatrix} = \begin{pmatrix} 1 \\ 0 \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} \begin{pmatrix} 0 \\ 1 \end{pmatrix} = \begin{pmatrix} 0 \\ i \end{pmatrix}$$

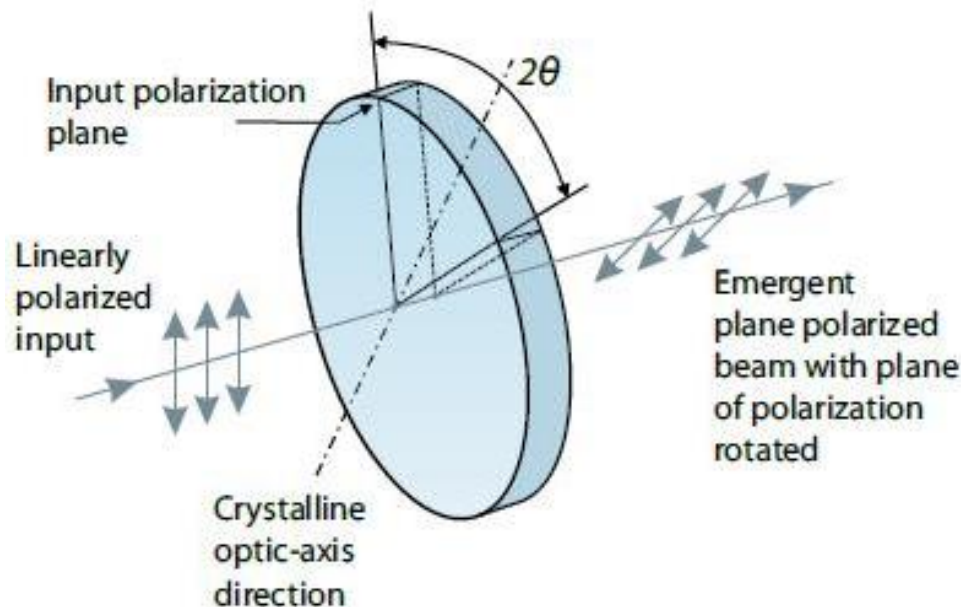
$$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} \begin{pmatrix} \cos \pi/4 \\ \sin \pi/4 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ i \end{pmatrix}$$

$$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi/2} \end{pmatrix} \begin{pmatrix} \cos \pi/4 \\ -\sin \pi/4 \end{pmatrix} = \frac{1}{\sqrt{2}} \begin{pmatrix} 1 \\ -i \end{pmatrix}$$

What about any θ ?

4.7 POLARIZATION

Matrix representation of polarizer operation



Performance of Half ($\lambda/2$) Waveplate

Half waveplate

$$M = \begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix}$$

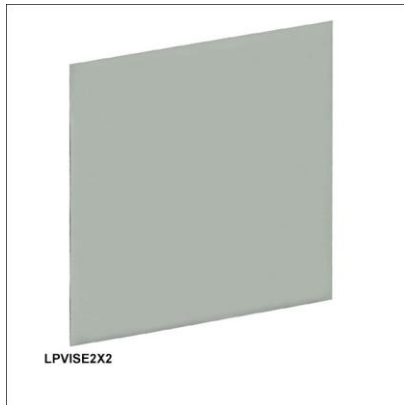
Retard vertical axis

$$\begin{pmatrix} 1 & 0 \\ 0 & e^{i\pi} \end{pmatrix} \begin{pmatrix} \cos \pi/4 \\ \sin \pi/4 \end{pmatrix} = \begin{pmatrix} \cos \pi/4 \\ -\sin \pi/4 \end{pmatrix}$$

What about any θ ?

4.8 POLARIZATION CREATION

Dichroic Polarizer (二向色薄膜偏振器) through anisotropic absorption



•Visible Dichroic Polarizer Extinction Ratios:

- $>100:1$ for 400 - 500 nm
- $>1000:1$ for 500 - 700 nm
- $>5000:1$ for 530 - 690 nm

2" x 2" Dichroic Film Polarizer Sheet

Triacetate Film, \$10



•NIR Diffuse Reflective Polarizers Extinction Ratios:

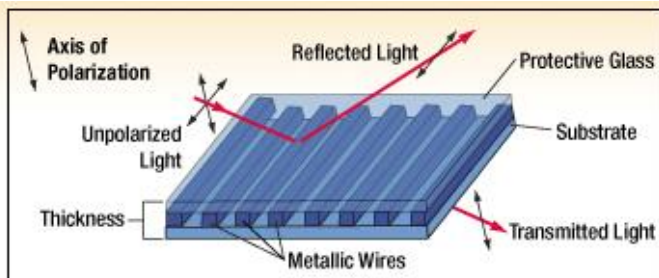
- $>400:1$ for 600 - 1100 nm
- $>1000:1$ for 600 - 940 nm

2" x 2" Diffuse Reflective Polarizer Sheet

Thermoplastic Polymer Film, \$100

4.8 POLARIZATION CREATION

Wire Grid Polarizers (线栅偏振片)



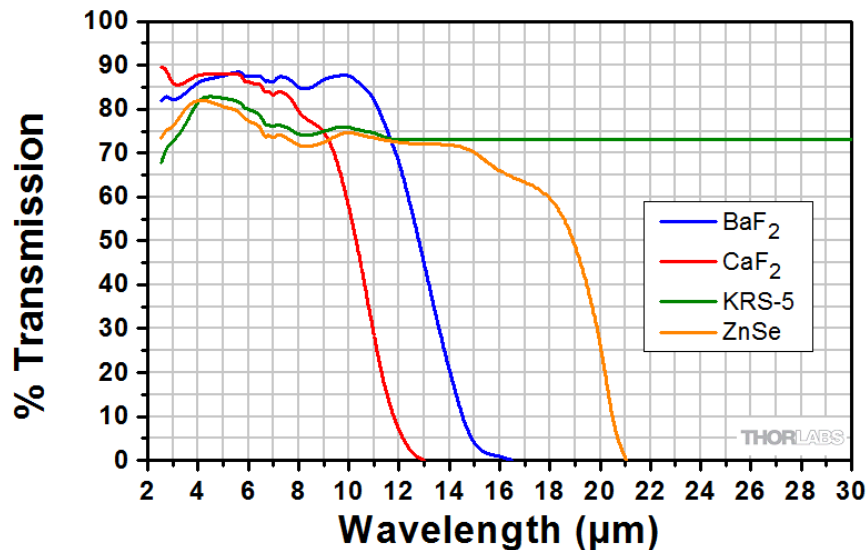
- Large Field of View
- Available Extinction Ratios
 - >800:1 (-VIS)
 - Up to 10 000:1 (-UB)
- Available Polarization Range:
 - 420 - 700 nm (-VIS)
 - 250 nm - 4 μm (-UB)

25.0 mm x 25.0 mm Wire Grid Polarizer, 250 nm to 4 μm , \$1200

4.8 POLARIZATION CREATION

Holographic Wire Grid Polarizers

Maximum Transmission



- Polarize Light from 2 - 30 μm
- Extinction Ratios from 150:1 to 300:1

KRS-5 Holographic Wire Grid Polarizer, Ø50 mm, Mounted, ~\$3500

4.8 POLARIZATION CREATION

Reflection from dielectric surfaces

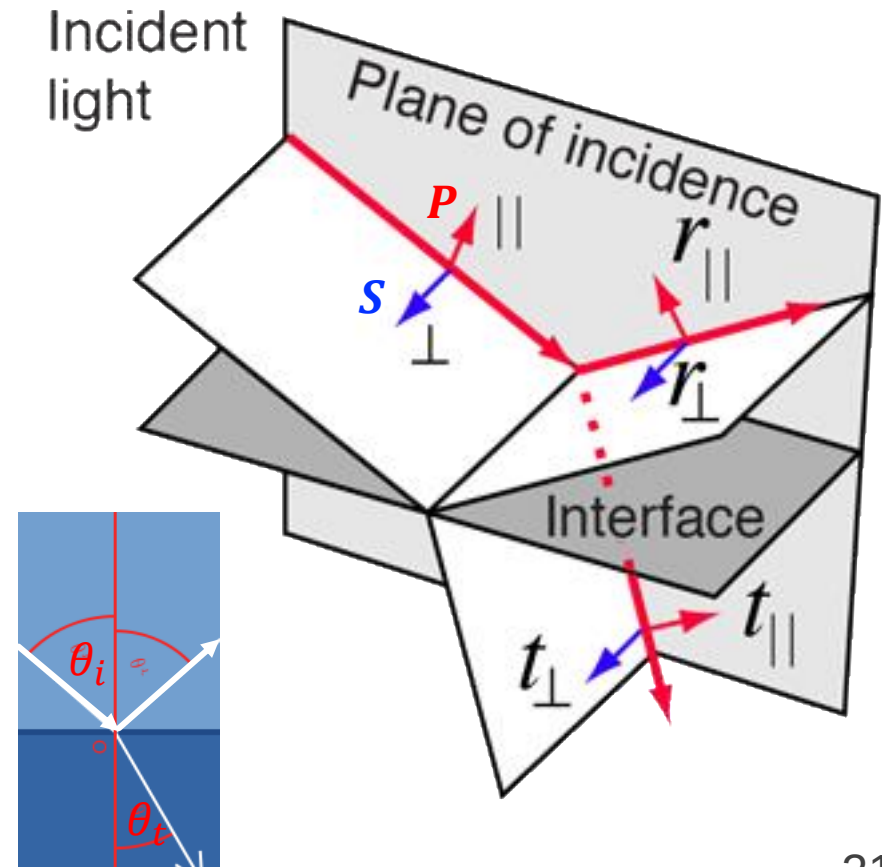
Snell's Law

$$\frac{\sin \theta_i}{\sin \theta_t} = \frac{n_2}{n_1}$$

Fresnel equations

$$R_s = \left[\frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} \right]^2$$

$$R_p = \left[\frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)} \right]^2$$



4.8 POLARIZATION CREATION

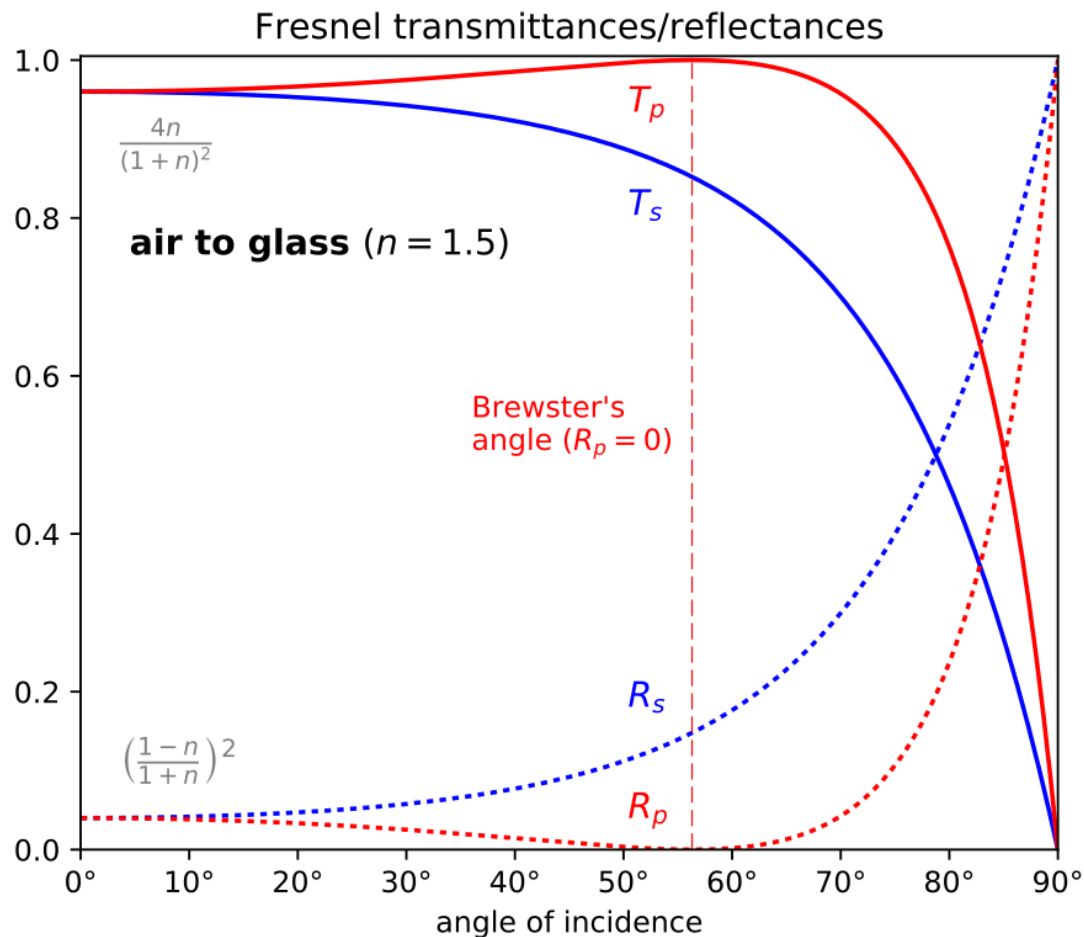
Reflection from dielectric surfaces

Fresnel equations

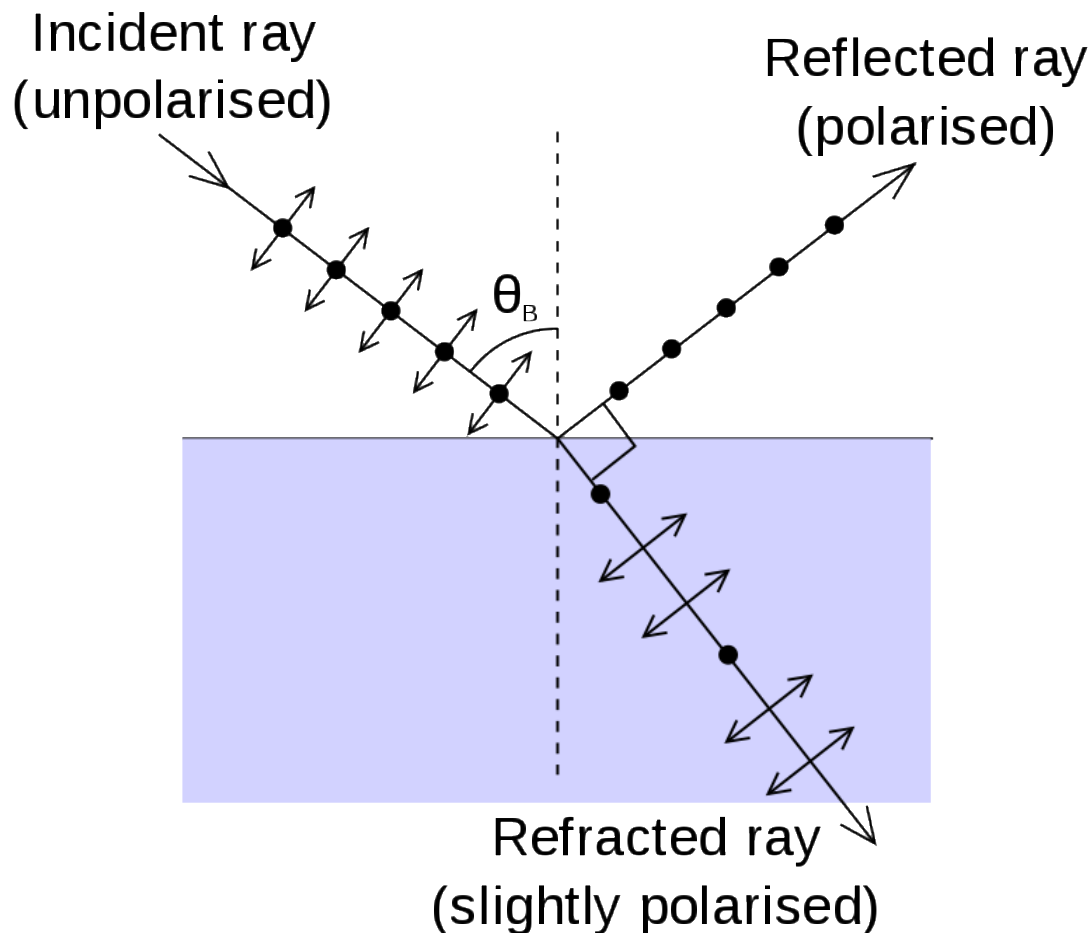
$$R_s = \left[\frac{\sin(\theta_t - \theta_i)}{\sin(\theta_t + \theta_i)} \right]^2 = \left(\frac{n_1 \cos \theta_i - n_2 \cos \theta_t}{n_1 \cos \theta_i + n_2 \cos \theta_t} \right)^2 = \left[\frac{n_1 \cos \theta_i - n_2 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i \right)^2}}{n_1 \cos \theta_i + n_2 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i \right)^2}} \right]^2$$

$$R_p = \left[\frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)} \right]^2 = \left(\frac{n_1 \cos \theta_t - n_2 \cos \theta_i}{n_1 \cos \theta_t + n_2 \cos \theta_i} \right)^2 = \left[\frac{n_1 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i \right)^2} - n_2 \cos \theta_i}{n_1 \sqrt{1 - \left(\frac{n_1}{n_2} \sin \theta_i \right)^2} + n_2 \cos \theta_i} \right]^2$$

4.8 POLARIZATION CREATION



4.8 POLARIZATION CREATION



Brewster angle

$$R_p = \left[\frac{\tan(\theta_t - \theta_i)}{\tan(\theta_t + \theta_i)} \right]^2$$

$$\theta_t + \theta_i = \frac{\pi}{2}$$

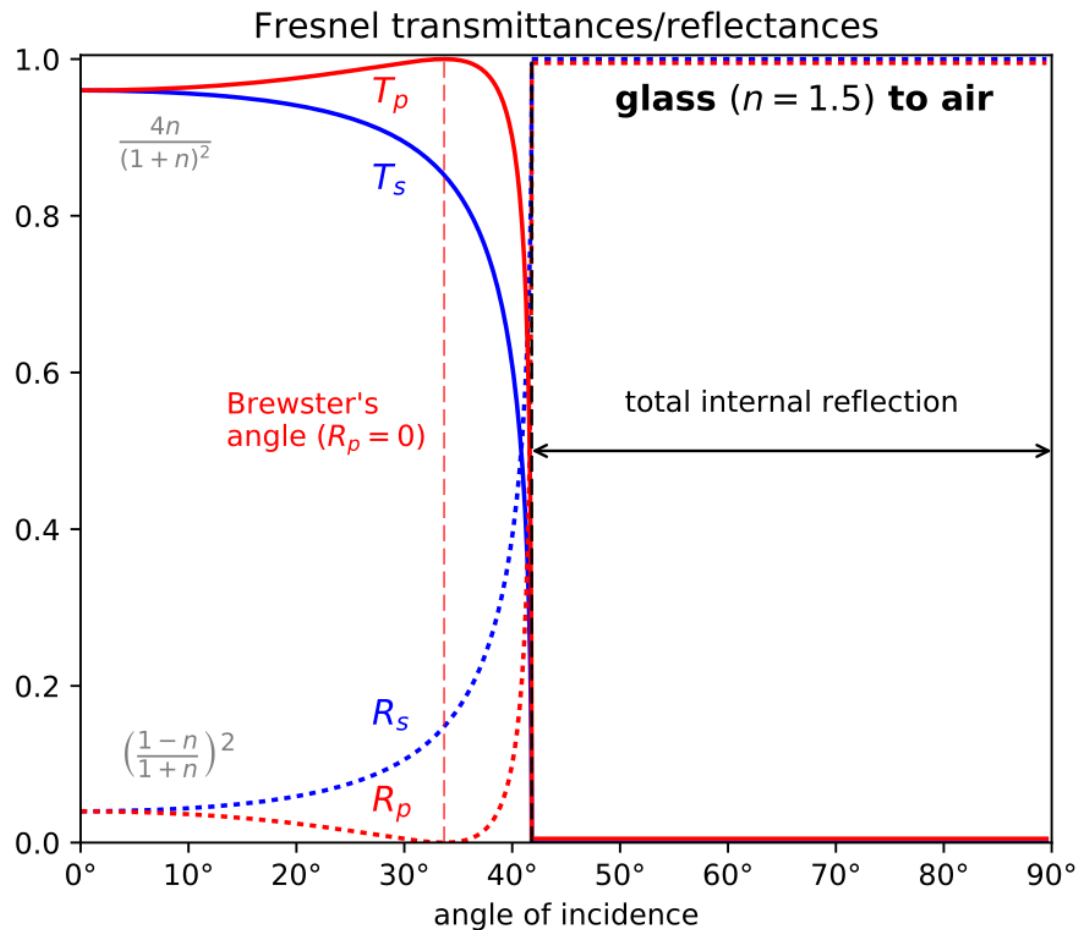
$$R_p = 0$$

$$n_1 \sin \theta_i = n_2 \sin \theta_t$$

$$\tan \theta_i = \frac{n_2}{n_1}$$

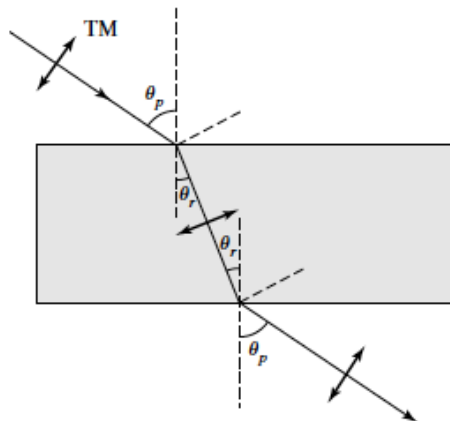
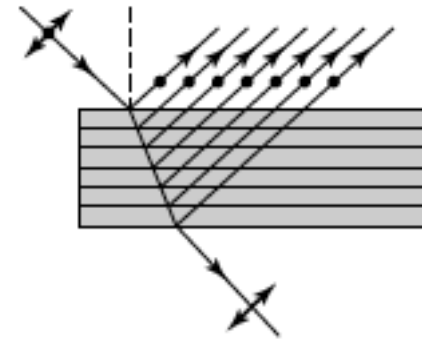
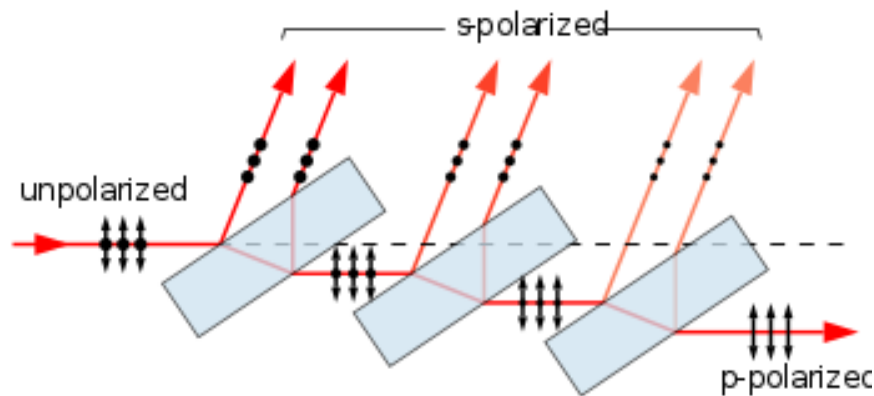
$$\theta_p = \arctan \frac{n_2}{n_1}$$

4.8 POLARIZATION CREATION



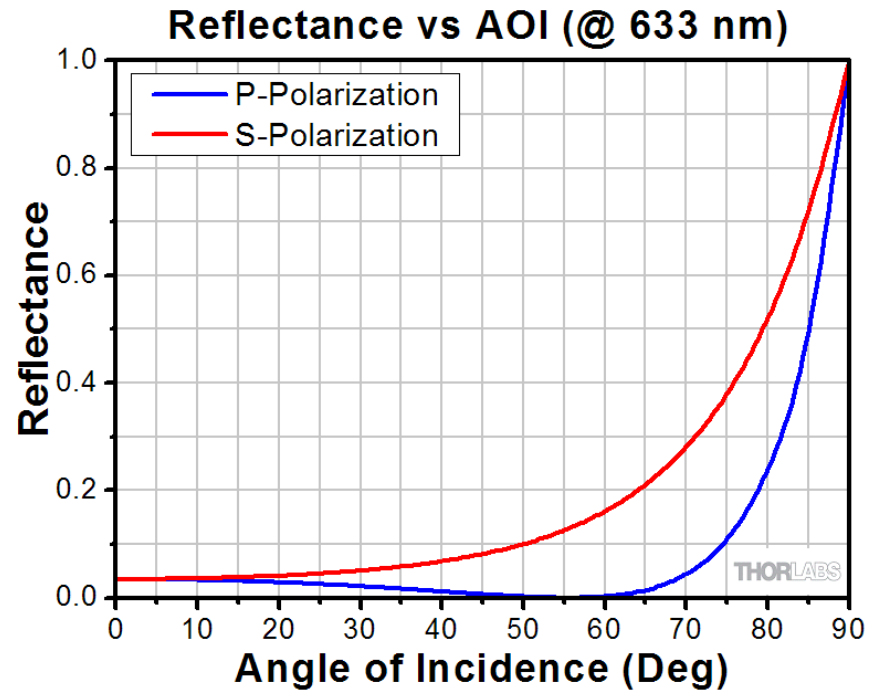
4.8 POLARIZATION CREATION

Pile-of-plates polarizer



fully transmitted

4.8 POLARIZATION CREATION



UVFS Brewster Window, 20.0 mm Minor Diameter, 2.0 mm Thick, ~\$ 200

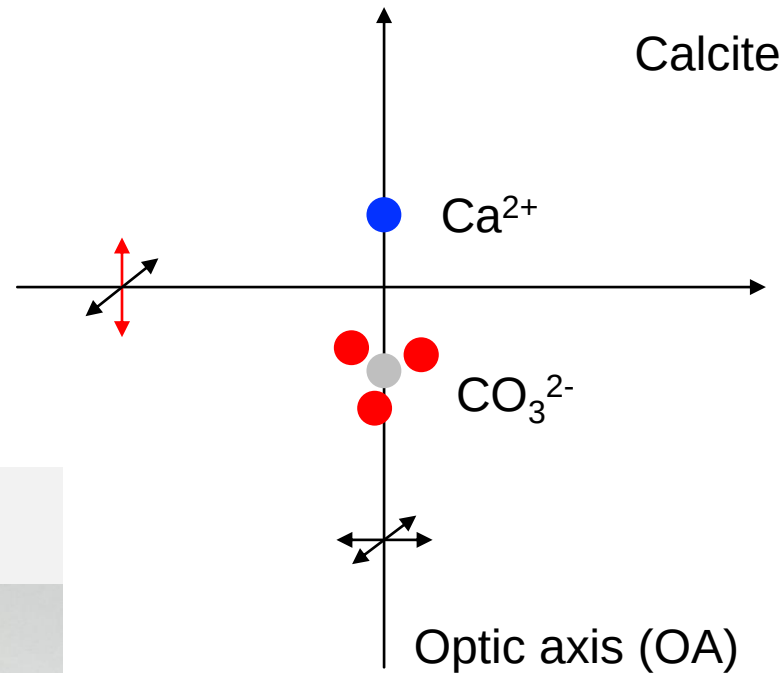
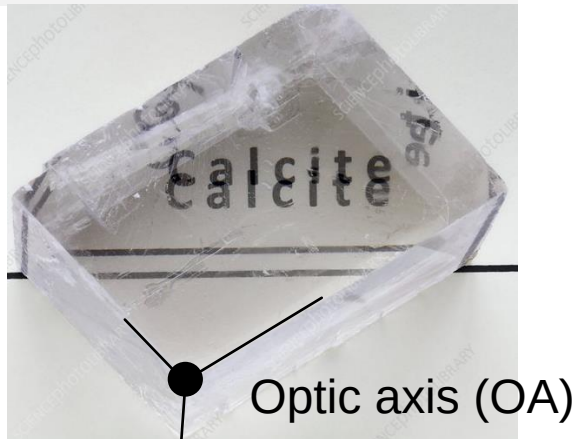
4.8 POLARIZATION CREATION

Birefringence

polarization $\perp$ OA

polarization $\parallel$ OA

Crystal	$n_{\perp}$	$n_{\parallel}$
<u>Calcite</u>	1.6584	1.4864

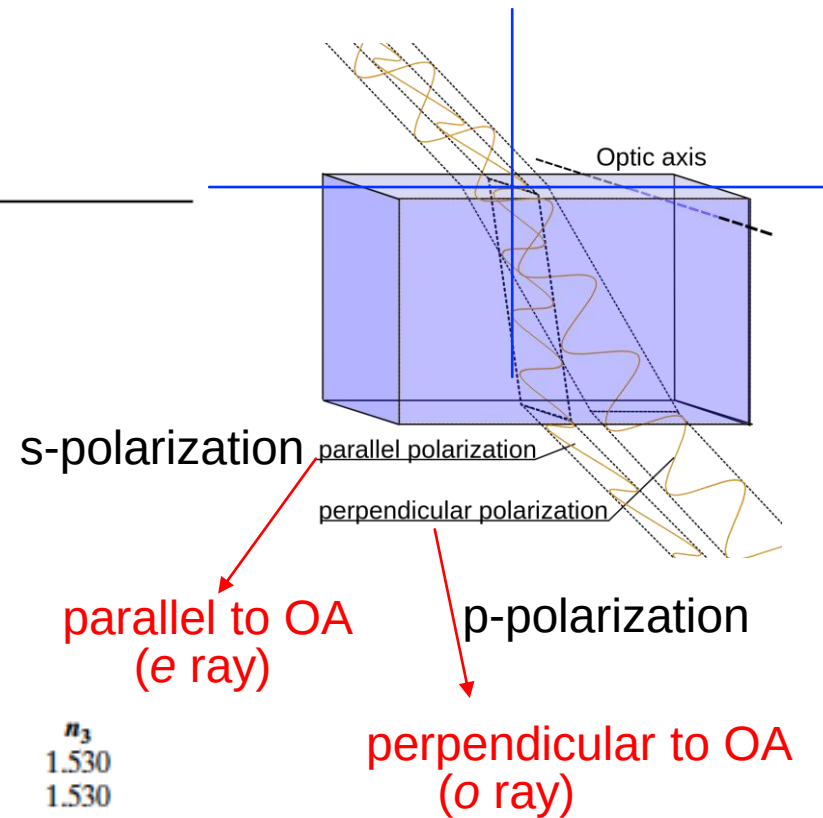


a direction in which a ray of transmitted light suffers no birefringence (ordinary)

4.8 POLARIZATION CREATION

Birefringence

Isotropic (cubic)	Sodium chloride	1.544		
	Diamond	2.417		
	Fluorite	1.392		
Uniaxial (trigonal, tetragonal, hexagonal)	Positive $n_{ } > n_{\perp}$:	$n_{ }$	$n_{\perp}$	
	Ice	1.313	1.309	
	Quartz (SiO_2)	1.5534	1.5443	
	Zircon (ZrSiO_4)	1.968	1.923	
	Rutile (TiO_2)	2.903	2.616	
	Negative $n_{ } < n_{\perp}$:			
	Calcite (CaCO_3)	1.4864	1.6584	
	Tourmaline	1.638	1.669	
	Sodium Nitrate	1.3369	1.5854	
	Beryl ($\text{Be}_3\text{Al}_2(\text{SiO}_3)_6$)	1.590	1.598	
Biaxial (triclinic, monoclinic, orthorhombic)		n_1	n_2	n_3
	Gypsum ($\text{CaSO}_4(2\text{H}_2\text{O})$)	1.520	1.523	1.530
	Feldspar	1.522	1.526	1.530
	Mica	1.552	1.582	1.588
	Topaz	1.619	1.620	1.627



4.8 POLARIZATION CREATION

Birefringence

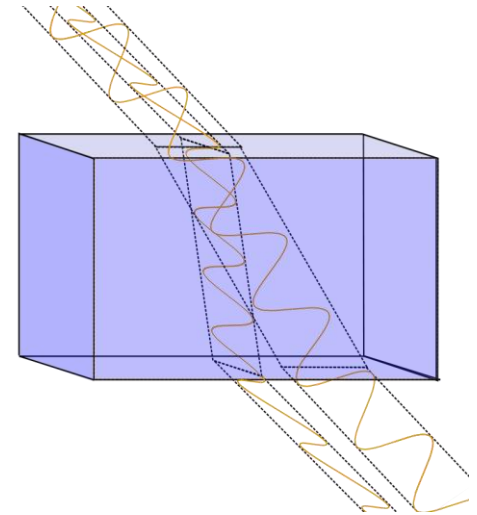
$$\delta = \frac{2\pi}{\lambda} |n_e - n_o| d$$

Quarter Wave Plate

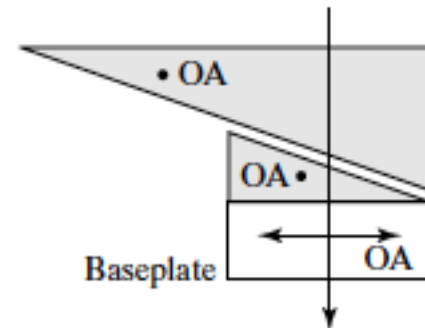
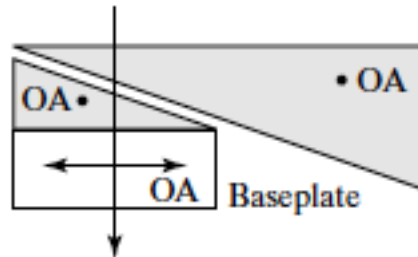
$$\delta = \frac{\pi}{2}$$

Half Wave Plate

$$\delta = \pi$$

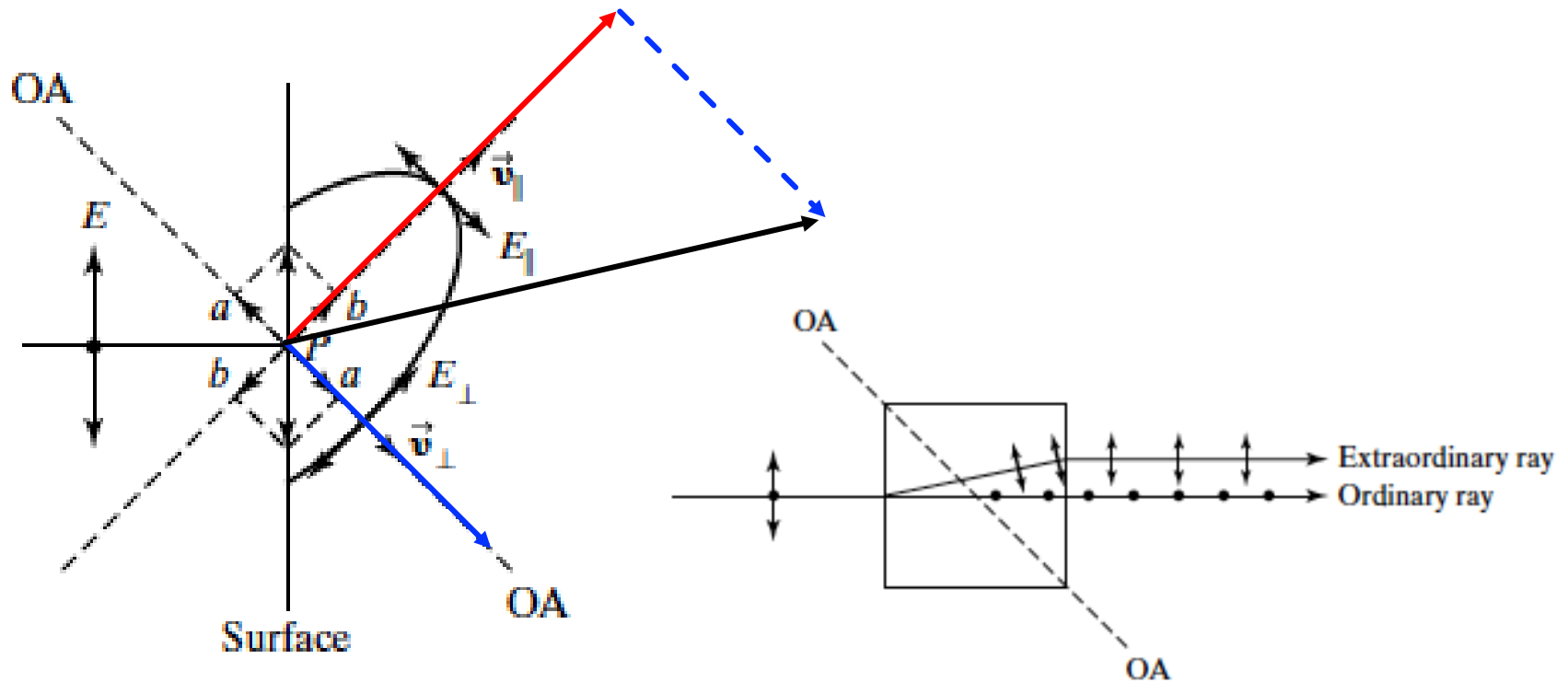


Soleil-Babinet compensator



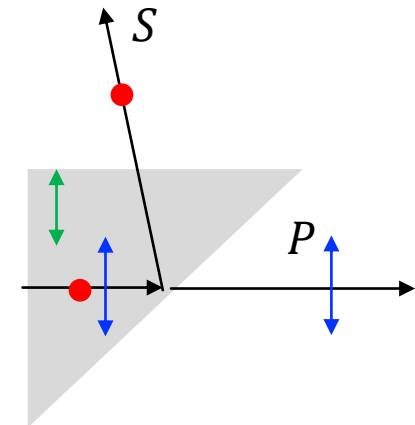
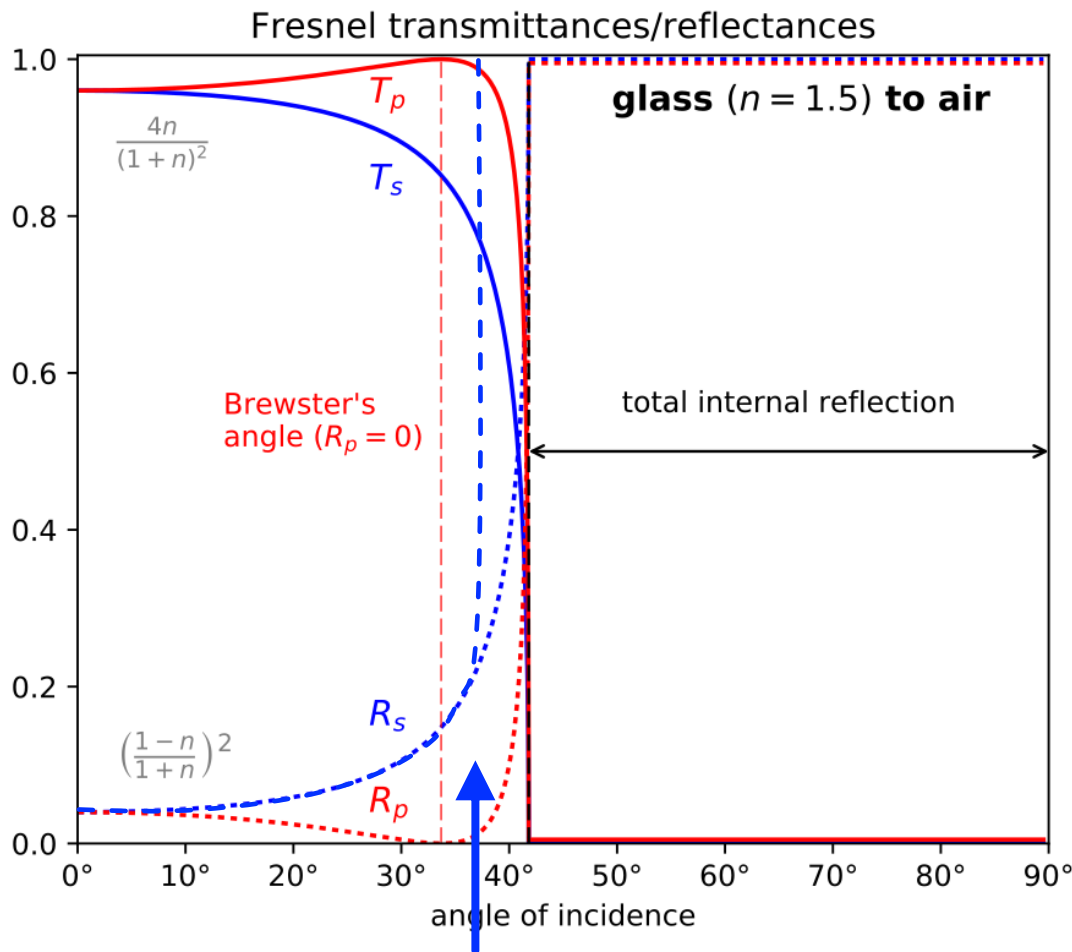
4.8 POLARIZATION CREATION

Birefringence



$$n_{\parallel} < n_{\perp}, \quad \frac{c}{n_{\parallel}} > \frac{c}{n_{\perp}}$$

4.8 POLARIZATION CREATION



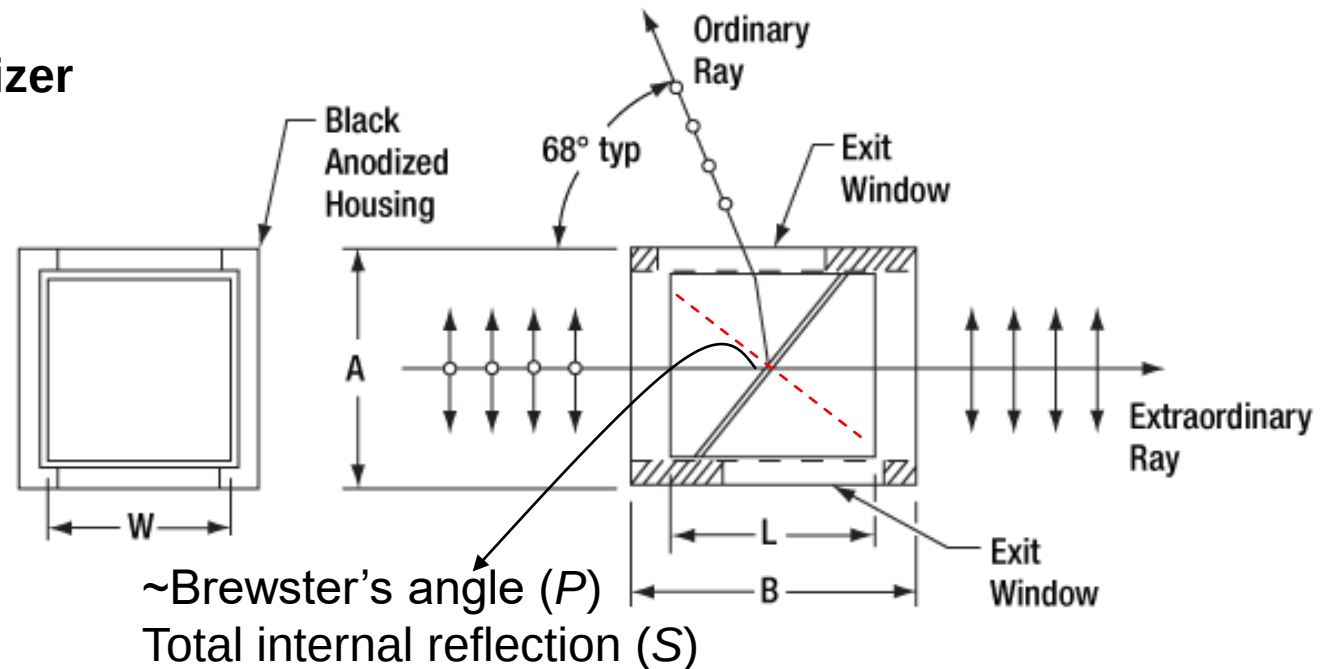
$$n_{\parallel} < n_{\perp}$$

$$n_p < n_s$$

4.8 POLARIZATION CREATION

Glan-Taylor Polarizer

Calcite



Extinction Ratio for Output Beam: 100 000:1

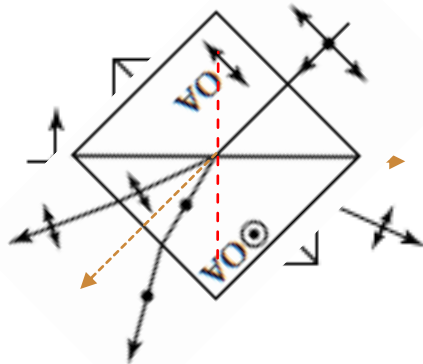
Glan-Taylor Design (Air-Spaced Birefringent Crystal Prisms)

Wavefront Distortion $\leq \lambda/4$ Over Clear Aperture

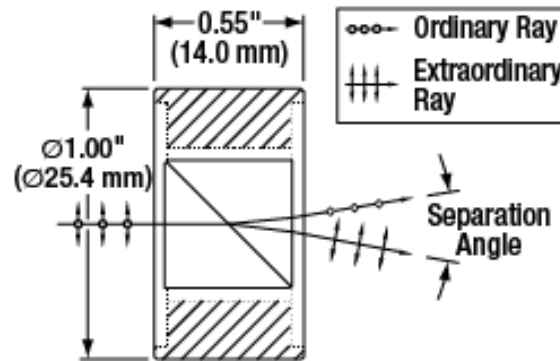
350 nm - 2.3 μm Wavelength Range (Uncoated)

4.8 POLARIZATION CREATION

Birefringence



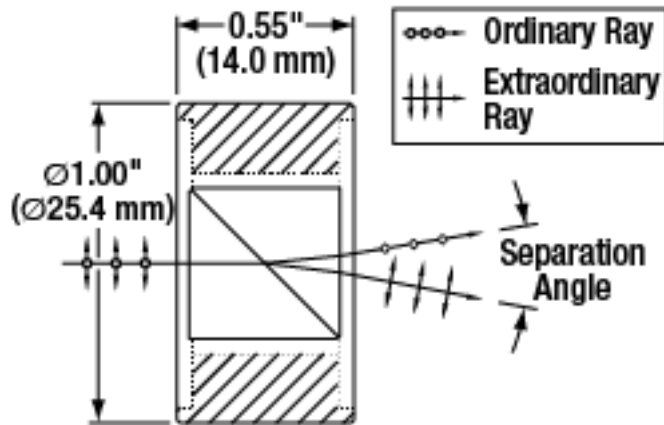
Wollaston



- Separates Unpolarized Light into Two Orthogonally Polarized Outputs
- High Extinction Ratio in Each Output
 - Quartz and MgF_2 : >10 000:1
 - α -BBO, YVO_4 , and Calcite: >100 000:1
- Prisms Available in Quartz, Magnesium Fluoride, α -BBO, or Calcite Substrates
- Calcite Prisms AR-Coated for 350 - 700 nm or 650 - 1050 nm

4.8 POLARIZATION CREATION

Wollaston Prisms



- Separates Unpolarized Light into Two Orthogonally Polarized Outputs
- High Extinction Ratio in Each Output
 - Quartz and MgF_2 : $>10\,000:1$
 - α -BBO, YVO_4 , and Calcite: $>100\,000:1$
- Prisms Available in Quartz, Magnesium Fluoride, α -BBO, or Calcite Substrates
- Calcite Prisms AR-Coated for 350 - 700 nm or 650 - 1050 nm

Wollaston Prism,, 190 nm - 3.5 μm , α -BBO, ~\$1200